Lake Gohlke — Team Captain

VyOS Router (VM 11) • Oversee All Teams

First 10 Minutes — VyOS Router (VM 11)

Min	Action
0–1	*// Connect to the router via SSH with default creds* `ssh vyos@172.16.101.1` (password: `changeme`). *// Enter configuration mode to make changes* `configure`. *// Immediately rotate the default password* `set system login user vyos authentication plaintext-password <NEWPASS>` then `commit`.
1–2	*// Run the automated hardening script (SSH lockdown, disable unused services, set firewall rules)* `bash harden_vm11_vyos.sh` — or paste commands manually from the script.
2–4	*// Confirm routing is intact and all subnets are reachable* `show ip route` — confirm routes to 172.16.101.0/24 (FW1), 172.16.102.0/24 (FW2), and default toward 172.31.21.0/29 (scoring).
4–5	*// Audit VyOS system users — there should typically only be the vyos account* `bash audit_vm11_vyos.sh`. Review output.
5–7	*// Test connectivity to both firewall outside interfaces* `ping 172.16.101.254` (FW1 outside), `ping 172.16.102.254` (FW2 outside). *// If pings fail, check interface status* `show interfaces`.
7–10	**Walk the room.** Confirm each team member is logged in and running their hardening scripts. Ensure Gray and Gavin are on the firewalls. Check with Imran on AD/DNS.

VyOS Command Reference

Configuration Mode

- *// Enter config mode* `configure`
- *// Apply pending changes* `commit`
- *// Write changes to disk (survive reboot)* `save`
- *// Leave config mode* `exit`
- *// Preview the current config tree* `show`
- *// Diff pending changes vs active config* `compare`
- *// Throw away uncommitted changes* `discard`
- *// Revert to a previous config revision* `rollback N`

Interfaces & Routing

- *// Display all interface IPs and link status* `show interfaces`
- *// Print the full routing table* `show ip route`
- *// Assign an IP to an interface* `set interfaces ethernet ethX address IP/MASK`
- *// Add a static route* `set protocols static route DEST/MASK next-hop GATEWAY`
- *// Remove a static route* `delete protocols static route DEST/MASK`
- *// Administratively disable an interface* `set interfaces ethernet ethX disable`

Firewall (Zone-Based)

- *// View all active firewall rulesets* `show firewall`
- *// Create/modify a rule (accept, drop, or reject traffic)* `set firewall name RULESET rule N action [accept|drop|reject]`
- *// Restrict rule to a source IP/subnet* `set firewall name RULESET rule N source address IP/MASK`
- *// Match on a destination port* `set firewall name RULESET rule N destination port PORT`
- *// Specify protocol (tcp, udp, icmp)* `set firewall name RULESET rule N protocol [tcp|udp|icmp]`
- *// Default policy: drop anything not explicitly allowed* `set firewall name RULESET default-action drop`
- *// Assign an interface to a zone* `set zone-policy zone NAME interface ethX`
- *// Apply a ruleset between zones* `set zone-policy zone NAME from OTHERZONE firewall name RULESET`

NAT

- *// Show current NAT translation table* `show nat rules`
- *// Set the inbound interface for destination NAT* `set nat destination rule N inbound-interface ethX`
- *// Translate destination to an internal server IP* `set nat destination rule N translation address INTERNAL_IP`
- *// Set the outbound interface for source NAT* `set nat source rule N outbound-interface ethX`
- *// Masquerade outbound traffic (PAT)* `set nat source rule N translation address masquerade`

Monitoring & Troubleshooting

- *// View system log entries* `show log`
- *// Live packet capture on an interface* `monitor traffic interface ethX`
- *// View the active connection tracking table* `show conntrack table ipv4`
- *// Check uptime and software version* `show system uptime` / `show version`
- *// View traffic flow stats (if NetFlow is configured)* `show flow-accounting`
- *// Restart DHCP if LAN clients can't get IPs* `restart dhcp server`

User Management

- *// List all configured system users* `show system login`
- *// Set or change a user's password* `set system login user USERNAME authentication plaintext-password PASS`
- *// Delete a user account* `delete system login user USERNAME`
- *// Change a user's privilege level* `set system login user USERNAME level [admin|operator]`

Captain Duties — Throughout Competition

1. **Stay off keyboard as much as possible.** Your job is coordination, not typing.
2. Keep a **status board**: which services are up/down, who is working what.
3. **Inject triage**: when injects arrive, immediately assign to Elissa (Inject Lead). Help prioritize.
4. **Score checks**: monitor the scoreboard. If a service drops, notify the responsible member immediately.
5. **Time awareness**: NISE portal is CST (we are EST = 1hr ahead). Inject deadlines are CST.
6. **Password ticket**: if a scored account password must change, file ticket at `auth.ccdc.events` FIRST.
7. **Red team IPs**: when an attacker IP is identified, tell Gray and Gavin to block the /32 on both firewalls immediately.
8. If someone is stuck >5 min on a problem, either help or reassign.

Red Team Activity Checklist (Every 30–60 Minutes)

1. *// Check for unknown users added to the router* `show system login` — look for any new accounts
2. *// Check if routing has been tampered with* `show ip route` — compare against the known good topology
3. *// Look for unexpected connections through the router* `show conntrack table ipv4` — look for anomalous flows
4. *// Verify firewall rules haven't been modified* `show firewall` — ensure no extra allow-all rules
5. *// Check config for unauthorized changes* `compare` (in config mode) — see if pending changes exist
6. **Walk the room**: Ask each member: "Any anomalies? Services up? Need help?"
7. Check the **scoreboard** — any services flipped red?
8. Check the **NISE inject portal** — any new injects since last check?

Imran Hussein — Windows Admin Lead

AD/DNS Server 2019 (VM 5) • Windows 11 Workstation (VM 8) • Windows Domain

First 10 Minutes — VM 5 AD/DNS + VM 8 Win11 Wks

Min	Action
0–1	*// Log into the AD Domain Controller* Log into VM 5 (172.20.240.102) as `administrator` / `!Password123`. Open PowerShell as Admin. *// Change the local admin password (no ticket needed for admin accounts)* `net user administrator <NEWPASS>`.
1–3	*// Run the automated hardening script (GPO, firewall, audit policy, disable unneeded services)* `.\harden_vm5_ad_dns.ps1`.
3–4	*// Audit all accounts; DO NOT delete unknowns — they may be white/black team* `.\audit_vm5_ad_dns.ps1`. Review every user account. **Do NOT delete or disable accounts you don't recognize** — they may be white/black team or scored. Only change passwords for accounts you confirm are team-controlled.
4–5	*// Verify DNS is functional by querying the local resolver* `nslookup google.com 127.0.0.1`. If DNS fails, check: *// Check if the DNS service process is running* `Get-Service dns` and *// List all DNS zones* `Get-DnsServerZone`.
5–7	*// Move to the Windows 11 workstation* Log into VM 8 (172.20.240.100) as `administrator` / `!Password123`. Change password. *// Harden and audit the workstation* Run `.\harden_vm8_win11_wks.ps1` then `.\audit_vm8_win11_wks.ps1`. Note: `UserOne:ChangeMe123` is also a default account on this box.
7–10	*// Open AD Users & Computers to review the entire domain* Back on VM 5: Open **Active Directory Users and Computers** (`dsa.msc`). Audit domain accounts. Check group memberships — especially Domain Admins, Enterprise Admins, Schema Admins. Document everything.

User Audit Decision Framework

When the audit script runs, for each account ask:

1. **Is it a scored service account?** (used for POP3 login, SMTP, etc.) → Change password via `auth.ccdc.events` ticket FIRST, then change it.
2. **Is it in Domain Admins but shouldn't be?** → Remove from group: *// Remove a user from the Domain Admins group* `Remove-ADGroupMember "Domain Admins" -Members USERNAME`
3. **Is it unknown/suspicious?** → **Disable** (don't delete): *// Disable the account to prevent login without destroying forensic data* `Disable-ADAccount -Identity USERNAME`
4. **Is it white/black team?** → **Do NOT touch.**

Windows Cheat Sheet — All Windows Boxes

Service Management

- *// List all currently running services* `Get-Service | Where Status -eq Running`
- *// Restart / start / stop a specific service* `Restart-Service <n>` / `Start-Service` / `Stop-Service`
- *// Open the graphical service manager* `services.msc`
- *// Query or disable a service via sc.exe* `sc query <service>` / `sc config <service> start= disabled`

Active Directory (VM 5)

- *// Open the AD Users & Computers GUI* `dsa.msc`
- *// List all AD users with their last logon date* `Get-ADUser -Filter * -Properties LastLogonDate | Select Name,Enabled,LastLogonD`
- *// Show who has Domain Admin privileges* `Get-ADGroupMember "Domain Admins"`
- *// Show group memberships for every user* `Get-ADUser -Filter * -Properties MemberOf | Select Name,MemberOf`
- *// Change an AD user's password* `Set-ADAccountPassword -Identity USER -NewPassword (ConvertTo-SecureString "PASS" -AsPlai`
- *// Disable an AD account* `Disable-ADAccount -Identity USERNAME`

- *// Find and unlock any locked-out accounts (e.g. scored users locked by brute force)* `Search-ADAccount -LockedOut | Unlock-ADAccount`

DNS Troubleshooting (VM 5 — Scored)

- *// Check if the DNS service is running* `Get-Service dns`
- *// Restart the DNS service* `Restart-Service dns`
- *// List all DNS zones hosted on this server* `Get-DnsServerZone`
- *// View all records in a specific zone* `Get-DnsServerResourceRecord -ZoneName "ZONE"`
- *// Test DNS resolution locally* `nslookup HOSTNAME 127.0.0.1`
- *// Open the DNS Manager GUI* `dnsmgmt.msc`
- *// Flush the server-side DNS cache* `Clear-DnsServerCache`
- *// Verify DNS is listening on port 53* `netstat -an | findstr :53`

Firewall

- *// Check if all firewall profiles (Domain, Private, Public) are enabled* `Get-NetFirewallProfile`
- *// Enable all three firewall profiles* `Set-NetFirewallProfile -All -Enabled True`
- *// List all active firewall rules* `Get-NetFirewallRule | Where Enabled -eq True | Select DisplayName,Direction,Action`
- *// Block a specific IP address at the host firewall* `New-NetFirewallRule -DisplayName "Block X" -Direction Inbound -RemoteAddress`
- *// Open the Windows Firewall GUI with advanced security* `wf.msc`

Intrusion Detection

- *// Open Event Viewer to inspect security logs* `eventvwr.msc` — Check Security log for 4624 (logon), 4625 (failed), 4720 (account created), 4732 (added to group).
- *// Pull the last 50 security events filtered to logon/account events* `Get-WinEvent -LogName Security -MaxEvents 50 | Where Id -in 462`
- *// Show who is logged in via RDP/console right now* `qwinsta`
- *// List all running processes with their owning user* `tasklist /v`
- *// Show active network connections with associated PIDs* `netstat -ano`
- *// Check for persistence in startup programs (Sysinternals)* `autoruns`
- *// List all scheduled tasks — red team often uses these for persistence* `schtasks /query /fo LIST /v`
- *// Check registry Run keys for auto-start programs* `reg query HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run`

Group Policy (Domain-Wide Hardening)

- *// Open the Group Policy Management Console* `gpmc.msc`
- *// Force an immediate group policy update on this machine* `gpupdate /force`
- *// See which GPOs are currently applied* `gpresult /r`
- Key GPO settings: password complexity, account lockout (5 attempts/30 min), audit logon events, restrict remote desktop

Red Team Activity Checklist (Every 30–60 Minutes)

1. *// Check for new or re-enabled AD accounts* `Get-ADUser -Filter * -Properties Enabled,WhenCreated | Where {$_.WhenCreated -gt (Get-Date).AddHours(-1)}`
2. *// Verify Domain Admins hasn't been modified* `Get-ADGroupMember "Domain Admins"` — compare against your documented list
3. *// Check for failed and suspicious logon events* `Get-WinEvent -LogName Security -MaxEvents 100 | Where {($_.Id -eq 4625) -or ($_.Id -eq 4720)}`
4. *// Look for new scheduled tasks (persistence)* `schtasks /query /fo LIST /v | findstr /i "TaskName"` — compare against baseline
5. *// Check active sessions on the DC* `qwinsta` — any unknown sessions?
6. *// Verify DNS is still operational* `nslookup google.com 127.0.0.1`
7. *// Check registry Run keys for new entries* `reg query HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run`
8. *// Scan for unexpected listening ports* `netstat -ano | findstr LISTENING` — compare against known services

Gray Groves — Firewall Lead

Palo Alto FW1/Linux (VM 9) • Reference for FW2 (VM 10)

First 10 Minutes — VM 9 Palo Alto Firewall 1

Min	Action
0–1	*// Connect to the Linux-side Palo Alto firewall* `ssh admin@172.20.242.150` (password: `Changeme123`). *// Immediately change the default admin password* `configure` then `set mgt-config users admin password`.
1–4	*// Apply all zone, address object, service object, and security rule configs* Paste the hardening commands from `harden_vm9_pa_fw1.txt`. Review each rule as you paste — understand what it does.
4–6	*// Apply management plane hardening (disable HTTP/Telnet/SNMP, set pw policy, harden SSH)* Paste the PAN-OS hardening commands from `maccdc2026_panos_harden.sh`.
6–7	*// Commit all changes to the running config* `commit` and wait for it to complete. If commit fails, read the error — usually a typo or missing object reference.
7–9	*// Verify the running security policy* `show running security-policy`. *// Test that traffic would match the correct rule* `test security-policy-match from OUTSIDE to INSIDE source 1.1.1.1 destination 172.20.242.30 protocol 6 por` — should match HTTP rule.
9–10	Coordinate with Gavin (FW2). Confirm his firewall is also configured. Both firewalls should have matching zone protection and hardening.

PAN-OS Command Reference

Modes

- *// Enter configuration mode* `configure`
- *// Apply pending configuration changes* `commit`
- *// Return to operational mode* `exit`
- *// Run operational commands from within config mode* `run` (e.g., `run show system info`)

Security Policy

- *// View the currently active (committed) security rules* `show running security-policy` — operational mode
- *// View configured (possibly uncommitted) security rules* `show security policy` — config mode
- *// Test which rule a specific traffic flow would match* `test security-policy-match from ZONE to ZONE source IP destination IP pro`
- To add a rule:
  - *// Set the source zone for the rule* `set rulebase security rules RULENAME from ZONE`
  - *// Set the destination zone* `set rulebase security rules RULENAME to ZONE`
  - *// Set source address (or 'any')* `set rulebase security rules RULENAME source [any|ADDR_OBJ]`
  - *// Set destination address* `set rulebase security rules RULENAME destination [any|ADDR_OBJ]`
  - *// Set application filter* `set rulebase security rules RULENAME application [any|APP]`
  - *// Set service/port filter* `set rulebase security rules RULENAME service [any|SVC_OBJ]`
  - *// Set the action (allow or deny)* `set rulebase security rules RULENAME action [allow|deny]`
  - *// Enable end-of-session logging* `set rulebase security rules RULENAME log-end yes`
- *// Move a rule to the top (critical for block rules to take precedence)* `move rulebase security rules RULENAME top`
- *// Delete a rule entirely* `delete rulebase security rules RULENAME`

Block a Red Team IP (Do This Fast!)

Only block /32 addresses — subnet blocks violate competition rules.

1. *// Enter config mode* `configure`
2. *// Create an address object for the attacker IP* `set address RT-IP-X ip-netmask ATTACKER_IP/32 description "RT block HH:MM"`

3. *// Create a deny rule referencing the address object* `set rulebase security rules BLOCK-RT-X from OUTSIDE to INSIDE source RT-IP`
4. *// Move the block rule to the top so it evaluates first* `move rulebase security rules BLOCK-RT-X top`
5. *// Commit the block* `commit`

Do this on BOTH firewalls (tell Gavin to mirror on FW2).

Address & Service Objects

- *// List all defined address objects* `show address all`
- *// Create a new address object* `set address NAME ip-netmask IP/MASK description "DESC"`
- *// List all defined service objects* `show service all`
- *// Create a new service object (e.g. custom port)* `set service NAME protocol tcp port PORT`

Monitoring & Logs

- *// View all active sessions passing through the firewall* `show session all`
- *// Filter sessions from a specific source IP* `show session all filter source IP`
- *// Get detailed info on a specific session* `show session id NUM`
- *// View global traffic counters* `show counter global`
- *// Tail the authentication log in real time* `tail follow yes mp-log authd.log`
- *// View traffic logs* `show log traffic`
- *// View threat/IPS logs (if licensed)* `show log threat`
- *// Check firewall uptime and version info* `show system info`
- *// View all interface status* `show interface all`
- *// View the ARP table* `show arp all`
- *// View the routing table* `show routing route`

Troubleshooting Connectivity

- *// Ping from the firewall itself (specify source interface)* `ping source INTERFACE_IP host DEST_IP`
- *// Check if traffic for a service is reaching the firewall* `show session all filter destination PORT`
- If a scored service is down: (1) check the rule exists in `show running security-policy`, (2) check sessions: `show session all filt`
  (3) check the server itself is up
- *// Advanced packet debugging (use carefully — verbose output)* `debug dataplane packet-diag`

FW1 (VM 9) Specifics — Linux Zone

Server	IP	Scored Services
Ubuntu Ecom	172.20.242.30	HTTP (80), HTTPS (443)
Fedora Webmail	172.20.242.40	SMTP (25), POP3 (110), HTTP (80), HTTPS (443)
Splunk	172.20.242.20	None (internal logging)

FW2 (VM 10) Specifics — Windows Zone

Server	IP	Scored Services
AD/DNS	172.20.240.102	DNS (53 TCP/UDP)
Web Server	172.20.240.101	HTTP (80), HTTPS (443)
FTP Server	172.20.240.104	FTP (21), TFTP (69), NTP (123)

Red Team Activity Checklist (Every 30–60 Minutes)

1. *// Review active sessions for suspicious source IPs* `show session all` — look for high session counts from unknown IPs
2. *// Check traffic logs for denied connections (red team probing)* `show log traffic` — look for repeated denies from the same source
3. *// Verify security policy hasn't been modified* `show running security-policy` — compare against your expected ruleset
4. *// Check that no new admin users were added* `show admins all` — should only be the `admin` account
5. *// Verify management interface is not exposed* `show system info` — check management interface config
6. *// Check for sessions to unusual destination ports (C2 callbacks)* `show session all filter destination-port 4444` (and other common C2 ports: 1234, 8080, 8443, 9001)
7. Coordinate with Gavin — are both firewalls still in sync?

Gavin Workman — Windows Admin

Palo Alto FW2/Windows (VM 10) • Windows Web Server (VM 6)

First 10 Minutes — VM 10 Firewall then VM 6 Web Server

Min	Action
0–1	*// Connect to the Windows-side Palo Alto firewall* `ssh admin@172.20.240.200` (password: `Changeme123`). *// Change the default admin password immediately* `configure` then `set mgt-config users admin password`.
1–4	*// Apply zone, address object, and security rule configs for the Windows zone* Paste the firewall config from `harden_vm10_pa_fw2.txt`. *// Then apply management plane hardening* Paste PAN-OS hardening from `maccdc2026_panos_harden.sh`.
4–5	*// Commit all changes and verify with a policy test* `commit`. Verify: `test security-policy-match from OUTSIDE to INSIDE source 1.1.1.1 destination 172.20.240.101 protocol 6 port 8` — should match HTTP rule for web server.
5–6	*// Switch to the Windows web server* Log into VM 6 (172.20.240.101) as `administrator` / `!Password123`. Open PowerShell as Admin. *// Change admin password (no ticket needed)* `net user administrator <NEWPASS>`.
6–8	*// Run the automated hardening script (firewall, security policies, IIS hardening)* `.\harden_vm6_win_web.ps1`.
8–9	*// Audit user accounts using the same decision framework as Imran's section* `.\audit_vm6_win_web.ps1`.
9–10	*// Verify the web server is responding* Open browser to `http://127.0.0.1` and `https://127.0.0.1`. If down: *// Check IIS service status* `Get-Service W3SVC` and *// Reset IIS if needed* `iisreset`.

Key Difference: Local vs. Domain

This is critical — know which context you're in!

Local Machine	Active Directory Domain
// List local user accounts `net user`	*// List all domain user accounts* `net user /domain`
// Show local Administrators group members `net localgroup administrators`	*// Show Domain Admins group members* `Get-ADGroupMember "Domain Admins"`
// Open Local Users & Groups GUI `lusrmgr.msc`	*// Open AD Users & Computers GUI (DC only)* `dsa.msc`
// Change a local user's password `net user USERNAME PASS`	*// Change a domain user's password* `Set-ADAccountPassword`
// Manage THIS machine's firewall Firewall: `wf.msc`	*// Push policy to ALL domain machines* GPO: `gpmc.msc`

Rule of thumb: VM 6 (Web Server) is domain-joined. Local admin changes affect only VM 6. Domain changes (on VM 5) affect all Windows boxes. When in doubt, ask Imran.

Windows Cheat Sheet

IIS Web Server (VM 6 — Scored: HTTP/HTTPS)

- *// Check if IIS World Wide Web service is running* `Get-Service W3SVC`
- *// Start IIS or do a full reset* `Start-Service W3SVC` / `iisreset`
- *// Open the IIS Manager GUI* `inetmgr`
- *// Default web content directory* Default web root: `C:\inetpub\wwwroot\`
- Check bindings: IIS Manager → Sites → Default Web Site → Bindings (should have port 80 and 443)
- Check site is started: IIS Manager → Sites → right-click → Start
- *// Test web server locally from PowerShell* `Invoke-WebRequest http://localhost`

- *// IIS log directory* Logs: `C:\inetpub\logs\LogFiles\`
- **Do NOT modify web content** unless an inject requires it — the scoring engine checks for specific content

Service Troubleshooting

- *// List all stopped services* `Get-Service | Where Status -eq Stopped`
- *// Restart a specific service by name* `Restart-Service <n>`
- *// Open the graphical service manager* `services.msc`
- *// Check if something is listening on an expected port* `netstat -ano | findstr :PORT`
- *// Test a local TCP port connection* `Test-NetConnection -ComputerName localhost -Port 80`

Firewall (Local)

- *// Check all firewall profile states* `Get-NetFirewallProfile`
- *// Enable all firewall profiles* `Set-NetFirewallProfile -All -Enabled True`
- *// Block a specific IP at the host firewall* `New-NetFirewallRule -DisplayName "Block IP" -Direction Inbound -RemoteAddress IP -A`
- *// List enabled firewall rules* `Get-NetFirewallRule | Where Enabled -eq True | ft DisplayName,Direction,Action`

Intrusion Detection

- *// Open Event Viewer; check Security log for key event IDs* `eventvwr.msc` — 4624=logon, 4625=failed, 4720=new account, 4732=group change
- *// Show active RDP/console sessions* `qwinsta`
- *// List processes with owning user context* `tasklist /v`
- *// Show all connections and their owning PIDs* `netstat -ano` — cross-reference suspicious PIDs with `tasklist`
- *// Check for persistence via scheduled tasks* `schtasks /query /fo LIST /v`
- *// Check Run key for auto-start programs* `reg query HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run`
- *// Pull recent security events for logon and account activity* `Get-WinEvent -LogName Security -MaxEvents 100 | Where {($_.Id -eq 4`

Palo Alto Quick Reference (FW2 — VM 10)

Same commands as Gray's section. Key details for your firewall:

- *// SSH to your firewall's management interface* Management: `ssh admin@172.20.240.200`
- Inside interface: 172.20.240.254/24 (Windows LAN)
- Outside interface: 172.16.102.254/24 (to VyOS)
- Protected servers: AD/DNS (.102), Web (.101), FTP (.104), Win11 Wks (.100)

When Gray calls out a red team IP — block it on your firewall too! Use the same block procedure from Gray's section.

Red Team Activity Checklist (Every 30–60 Minutes)

1. *// Check FW2 active sessions for suspicious traffic* `show session all` (on FW2) — look for high session counts or unusual destination ports
2. *// Verify FW2 security policy is intact* `show running security-policy` — any unexpected allow-all rules?
3. *// On VM6: check for new local accounts* `net user` — compare against your documented list
4. *// On VM6: check active sessions* `qwinsta` — unknown sessions?
5. *// Verify IIS is still serving content correctly* `Invoke-WebRequest http://localhost` — is the expected content returned?
6. *// Check for new scheduled tasks on VM6* `schtasks /query /fo LIST /v | findstr /i "TaskName"`
7. *// Verify Windows firewall is still enabled on VM6* `Get-NetFirewallProfile`
8. Mirror any red team IP blocks from Gray onto FW2

Elissa Thomas — Inject Lead

Windows FTP Server 2022 (VM 7) • Inject Management • IR Reporting

First 10 Minutes — VM 7 FTP Server

Min	Action
0–1	*// Log into the FTP server with default credentials* Log into VM 7 (172.20.240.104) as `administrator` / `!Password123`. Open PowerShell as Admin. *// Change admin password (no ticket needed)* `net user administrator <NEWPASS>`.
1–3	*// Run the hardening script (firewall, FTP/TFTP/NTP hardening, disable unneeded services)* `.\harden_vm7_ftp.ps1`.
3–5	*// Audit users — FTP service accounts must stay active for scoring* `.\audit_vm7_ftp.ps1`. Review user accounts. **Do NOT delete FTP service accounts.** Disable suspicious accounts, remove unnecessary admin group memberships.
5–7	*// Verify FTP works by connecting locally* `ftp localhost` — log in with a known account. *// Check if TFTP service is running* `Get-Service` (search for TFTP). *// Verify NTP is operational* `w32tm /query /status`.
7–8	Check FTP passive port range. If configurable, narrow it (e.g., 60000–65535) and tell Gavin to update the firewall rule.
8–10	Open the NISE portal in a browser. Familiarize yourself with the inject submission interface. **Remember: NISE times are CST. We are EST = 1 hour ahead.**

FTP/TFTP/NTP Troubleshooting (All Scored on VM 7)

FTP (Port 21 — Scored)

- *// Check if the FTP service is running* `Get-Service MSFTPSVC` or `Get-Service ftpsvc`
- *// Start the FTP service or use IIS Manager* `Start-Service ftpsvc` / IIS Manager → FTP site → Start
- *// Verify FTP is listening on port 21* `netstat -an | findstr :21`
- *// Open IIS Manager to check FTP bindings and auth settings* `inetmgr` → FTP site → check bindings, authentication, authorization
- FTP root directory: check IIS for the physical path
- *// Test FTP locally* `ftp localhost` — can you log in and list files?
- Common fix: if FTP auth fails, check IIS FTP Authentication settings (Basic Auth must be enabled)

TFTP (Port 69/UDP — Scored)

- *// TFTP may be a Windows Feature or third-party — find the service name* `Get-Service | Where DisplayName -like "*tftp*"`
- *// Verify TFTP is listening on UDP port 69* `netstat -an | findstr :69`
- TFTP root directory: check service configuration for the file path
- Scoring checks file retrieval with integrity — make sure files exist in the TFTP root

NTP (Port 123/UDP — Scored)

- *// Check NTP synchronization status* `w32tm /query /status`
- *// View NTP configuration details* `w32tm /query /configuration`
- *// Check if the Windows Time service is running* `Get-Service W32Time` / `Start-Service W32Time`
- *// Configure NTP peers and force a manual sync* `w32tm /config /manualpeerlist:"pool.ntp.org" /syncfromflags:manual /update`
- *// Force an immediate time resync* `w32tm /resync`
- *// Verify NTP is listening on UDP port 123* `netstat -an | findstr :123`
- *// NTP registry settings location* Registry: `HKLM\SYSTEM\CurrentControlSet\Services\W32Time\Parameters`

Inject Best Practices

1. **Read the entire inject before starting.** Understand exactly what is asked.

2. **Check the deadline** (NISE is CST, we are EST — subtract 1 hour from our clocks to get NISE time).
3. **Delegate technical work.** If the inject requires Linux config, hand it to Wesley/Nathaniel. Windows config goes to Imran/Gavin.
4. **Submit as PDF** via NISE. Use a clean, professional format.
5. **Partial credit exists.** If time is running out, submit what you have — something is better than nothing.
6. **Prioritize by points.** Ask Lake which injects are highest value if multiple arrive at once.
7. **Document everything.** Screenshots, commands run, results — include in the PDF.

IR Report Template

When an incident is detected, use this format (from the scoring rubric):

1. **Affected Host:** hostname, IP, OS
2. **Source IP / Dest IP:** attacker → victim
3. **Port/Service:** what was targeted
4. **Timestamps:** when detected, when it started
5. **Affected Account:** which user was compromised
6. **Summary:** 2–3 sentence overview
7. **Vulnerability:** what was exploited + evidence (logs, screenshots)
8. **Initial Access:** how they got in + evidence
9. **Impact:** what damage was done + evidence
10. **Eradication:** how you removed the threat + evidence
11. **Remediation:** what you did to prevent recurrence + evidence

Every section needs EVIDENCE — screenshots, log snippets, command output.

Red Team Activity Checklist (Every 30–60 Minutes)

1. *// Verify FTP is still responding* `ftp localhost` — can you still log in?
2. *// Check if TFTP files are intact* Verify expected files still exist in the TFTP root directory
3. *// Verify NTP is still syncing* `w32tm /query /status`
4. *// Check for new local accounts on VM7* `net user` — compare against your documented list
5. *// Check active sessions* `qwinsta` — anyone unexpected logged in?
6. *// Check for scheduled task persistence* `schtasks /query /fo LIST /v | findstr /i "TaskName"`
7. *// Check for suspicious connections* `netstat -ano | findstr ESTABLISHED` — unknown IPs?
8. Check NISE portal for **new injects**

Liv Litaker — IR Lead / FLEX

Ubuntu Workstation (VM 4) • Incident Response • Flex Support

First 10 Minutes — VM 4 Ubuntu Workstation

Min	Action
0–1	*// Log in with default credentials* Log into VM 4 as `sysadmin` / `changeme`. Open terminal. *// Change the sysadmin password* `passwd`. *// Escalate to root and change the root password* `sudo -i` and `echo "root:<NEWPASS>" | chpasswd`.
1–3	*// Run the hardening script (creates admin user, locks root, hardens SSH, sets iptables OUTPUT DROP, installs fail2ban)* `sudo bash harden_vm4_ubuntu_wks.sh <NEWPASS> changeme`.
3–5	*// Audit users; this is a workstation so it should have very few accounts* `sudo bash audit_vm4_ubuntu_wks.sh`. Review users. **Cannot re-task this workstation** (competition rule).
5–7	Verify the machine is functional as a workstation. Check that GUI desktop works, browser opens. *// Verify the workstation received a DHCP address* `ip a`.
7–10	*// Start monitoring: see what services are exposed* `sudo ss -tulpn`. *// Watch auth log in real time for login attempts* `sudo tail -f /var/log/auth.log` in a terminal. You are now in IR mode.

Linux Command Cheat Sheet

User Management

- *// List all non-system (human) user accounts* `cat /etc/passwd | awk -F: '$3 >= 1000'`
- *// Show last login time for every user* `lastlog`
- *// Show recent login history* `last`
- *// Show who is currently logged in* `who` / `w`
- *// Change a user's password* `passwd USERNAME`
- *// Lock (disable) an account* `usermod -L USERNAME`
- *// Delete a user and their home directory* `userdel -r USERNAME`
- *// Show what groups a user belongs to* `groups USERNAME`
- *// Remove a user from the sudo group* `deluser USERNAME sudo`
- *// Show UID, GID, and all groups for a user* `id USERNAME`

Service Management

- *// Check if a service is running* `systemctl status SERVICE`
- *// Restart / start / stop a service* `systemctl restart SERVICE` / `start` / `stop`
- *// List all currently running services* `systemctl list-units -type=service -state=running`
- *// Prevent a service from starting on boot* `systemctl disable SERVICE`
- *// Show all listening ports and their owning processes* `ss -tulpn`
- *// Follow live log output for a specific service* `journalctl -u SERVICE -f`

Network

- *// Show IP addresses assigned to all interfaces* `ip a`
- *// Show the routing table* `ip r`
- *// Show all listening ports* `ss -tulpn`
- *// Show all established TCP connections* `ss -antp`
- *// Live packet capture on an interface* `tcpdump -i eth0 -n`
- *// Show connections with PIDs (if netstat is installed)* `netstat -antp`

Intrusion Detection

- *// Watch SSH and sudo login attempts in real time* `tail -f /var/log/auth.log`
- *// Search for brute force attempts* `grep "Failed password" /var/log/auth.log`
- *// Search for successful logins (look for unknown IPs)* `grep "Accepted" /var/log/auth.log`
- *// Show login history (look for unknown users/IPs)* `last`
- *// Check what commands were run as root* `cat /root/.bash_history`
- *// Find all bash_history files across all users* `find / -name ".bash_history" 2>/dev/null`
- *// Find suspicious executables in /tmp (common red team drop zone)* `find /tmp -executable -type f`
- *// Find all SUID binaries (compare against a known good list)* `find / -perm -4000 -type f 2>/dev/null`
- *// Check cron jobs for persistence* `crontab -l` / `ls /etc/cron*`
- *// Look for new accounts with UID 0 (root equivalents)* `cat /etc/passwd`
- *// Find system files modified in the last 24 hours* `find /etc /bin /sbin /usr -type f -mtime -1`
- *// Check for immutable file flags (red team trick to prevent file changes)* `lsattr /etc/`
- *// Find the top CPU-consuming processes (look for crypto miners)*

IR Reporting Best Practices

1. **Detect**: watch logs, watch connections, watch processes. When something is wrong, timestamp it.
2. **Document FIRST, act second.** Screenshot the evidence BEFORE you kill the process or remove the file.
3. **Collect evidence:** *// Dump running processes to a file* `ps aux > /root/evidence_ps.txt`, *// Dump network connections to a file* `ss -antp > /root/evidence_net.txt`, screenshot logs.
4. **Use the IR template** (same as Elissa's section) for every report.
5. **Every field needs evidence.** Logs, screenshots, command output — attach to the report.
6. **Coordinate with Lake** on which incidents to report and prioritize.
7. **After eradication:** verify the service is back up, document what you changed, monitor for re-entry.

IR Report Quick Template

Same format as Elissa's section: Affected Host, Source/Dest IPs, Port/Service, Timestamps, Affected Account, Summary, Vulnerability + evidence, Initial Access + evidence, Impact + evidence, Eradication + evidence, Remediation + evidence.

Red Team Activity Checklist (Every 30–60 Minutes)

1. *// Check for new or unauthorized users* `cat /etc/passwd | awk -F: '$3 >= 1000'` — any new entries?
2. *// Check for active sessions from unknown IPs* `who` and `last | head -20`
3. *// Look for suspicious processes*
3. *// Check for new cron jobs (persistence)* `crontab -l` and `for u in $(cut -f1 -d: /etc/passwd); do echo "-- $u --"; crontab`
4. *// Check for recently modified system files* `find /etc /bin /sbin /usr -type f -mmin -60`
5. *// Check for suspicious executables in temp dirs* `find /tmp /var/tmp /dev/shm -executable -type f`
6. *// Review auth log for recent activity* `tail -50 /var/log/auth.log` — any failed or unexpected logins?
7. *// Check outbound connections (should be blocked by iptables OUTPUT DROP)* `ss -antp | grep ESTAB` — anything unexpected?
8. Check with team members — anyone report anomalies on their boxes?

Wesley Heltzel — Linux Lead

Ubuntu Ecom (VM 1) • All Linux Systems Reference

First 10 Minutes — VM 1 Ubuntu Ecom

Min	Action
0–1	*// SSH into the Ecom server with default creds* `ssh sysadmin@172.20.242.30` (password: `changeme`). *// Change sysadmin password* `passwd`. *// Escalate to root and change root password* `sudo -i`, `echo "root:<NEWPASS>" | chpasswd`.
1–4	*// Run the hardening script (admin user, lock root, SSH hardening, iptables OUTPUT DROP, fail2ban, PHP hardening)* `sudo bash harden_vm1_ubuntu_ecom.sh <NEWPASS> changeme`.
4–6	*// Audit users — service accounts like mysql/postgres must NOT be deleted* `sudo bash audit_vm1_ubuntu_ecom.sh`. Review users carefully. **Do NOT delete service accounts.** Disable suspicious human accounts.
6–8	*// Verify the web server is responding on both HTTP and HTTPS* `curl -I http://localhost` and `curl -Ik https://localhost`. *// If down, check the web server service* `systemctl status apache2` (or nginx). *// Start it if stopped* `systemctl start apache2`.
8–9	Check what the scoring engine expects: browse to the site and note the content. **Do NOT modify web content** unless required by an inject.
9–10	*// If time permits, harden Splunk on VM3* Check Splunk (VM 3, 172.20.242.20). SSH in, verify: `systemctl status splunkd`. Change default creds (admin:changeme on web UI, root:changemenow on SSH). Run `harden_vm3_splunk.sh` if time permits.

Web Server Troubleshooting (HTTP/HTTPS — Scored)

Apache

- *// Check if Apache is running* `systemctl status apache2`
- *// Restart Apache* `systemctl restart apache2`
- *// Test config syntax before restarting (avoid breaking a working config)* `apache2ctl -t`
- *// Watch error log in real time* `tail -f /var/log/apache2/error.log`
- *// Watch access log for scoring checks and attacks* `tail -f /var/log/apache2/access.log`
- *// Main config and virtual host locations* Config: `/etc/apache2/apache2.conf`, sites: `/etc/apache2/sites-enabled/`
- *// Check the document root directory* Document root: typically `/var/www/html/` — verify in config
- *// Enable SSL module and default SSL site for HTTPS* `a2enmod ssl` then `a2ensite default-ssl`
- *// Verify the web server is bound to ports 80 and 443* `ss -tlnp | grep :80` / `grep :443`

Nginx (if used instead)

- *// Check/restart Nginx* `systemctl status nginx` / `systemctl restart nginx`
- *// Test Nginx config syntax* `nginx -t`
- *// Config and site locations* Config: `/etc/nginx/nginx.conf`, sites: `/etc/nginx/sites-enabled/`
- *// Log locations* Logs: `/var/log/nginx/error.log`, `/var/log/nginx/access.log`

E-Commerce App

- Check what framework (likely PHP-based: WordPress, Magento, PrestaShop, or custom)
- *// Check database service status* `systemctl status mysql` or `systemctl status postgresql`
- If the app is down but web server is up: check app logs, check database connectivity
- *// PHP config location — hardening script disables dangerous functions* `/etc/php/*/apache2/php.ini`

- **If the site content changes unexpectedly, red team may have defaced it.** *// Find web files modified in the last 24 hours* `find /var/www -mtime -1`

Linux Lead Cheat Sheet — All 4 Linux Boxes

Common Services Across Linux Boxes

VM	Host	Services	Key Ports
VM 1	Ubuntu Ecom	Apache/Nginx, PHP, DB	80, 443
VM 2	Fedora Webmail	Postfix, Dovecot, Roundcube	25, 110, 80, 443
VM 3	Splunk	Splunk Enterprise	8000 (web), 8089 (API), 9997 (forwarder)
VM 4	Ubuntu Wks	Desktop workstation	None scored

Package Management

- *// Update and install on Ubuntu/Debian* Ubuntu/Debian: `apt update && apt upgrade -y` / `apt install PACKAGE`
- *// Update and install on Fedora* Fedora: `dnf update -y` / `dnf install PACKAGE`
- *// Update and install on Oracle Linux / RHEL* Oracle Linux: `yum update -y` / `yum install PACKAGE`

iptables Quick Reference

- *// List all rules with packet counters* `iptables -L -n -v`
- *// Block a specific attacker IP* `iptables -A INPUT -s ATTACKER_IP -j DROP`
- *// Remove a block rule* `iptables -D INPUT -s ATTACKER_IP -j DROP`
- *// Open a specific port for inbound traffic* `iptables -A INPUT -p tcp -dport PORT -j ACCEPT`
- *// Save rules so they persist across reboots* `iptables-save > /etc/iptables/rules.v4`
- *// Default deny all outbound traffic (blocks red team reverse shells/C2)* `iptables -P OUTPUT DROP` — **critical for blocking red team callbacks**

Intrusion Detection (same as Liv's section)

`tail -f /var/log/auth.log`, `last`, `find /tmp -executable`, `crontab -l`, `ps aux`, `ss -antp`, `find / -perm -4000`, `lsattr /etc/`, `bash_history` for all users.

Red Team Activity Checklist (Every 30–60 Minutes)

1. *// Verify the Ecom web server is up and serving correct content* `curl -I http://localhost` and `curl -Ik https://localhost`
2. *// Check for web defacement* `find /var/www -mmin -60 -type f` — any recently modified files?
3. *// Check database is running* `systemctl status mysql` (or postgresql)
4. *// Check for new or unauthorized users* `cat /etc/passwd | awk -F: '$3 >= 1000'`
5. *// Check for suspicious processes*
5. *// Check for new cron jobs* `crontab -l` and `ls -la /etc/cron*`
6. *// Check iptables rules haven't been modified* `iptables -L -n -v` — is OUTPUT still DROP? Are rules intact?
7. *// Review auth log* `tail -50 /var/log/auth.log`
8. *// Check Splunk (VM3) is still running if you set it up* `ssh sysadmin@172.20.242.20 "systemctl status splunkd"`

Nathaniel Ahwee — Linux Admin

Fedora Webmail (VM 2) • Mail Services (SMTP/POP3 — Scored)

First 10 Minutes — VM 2 Fedora Webmail

Min	Action
0–1	*// SSH into the Fedora Webmail server with default creds* `ssh sysadmin@172.20.242.40` (password: `changeme`). *// Change sysadmin password* `passwd`. *// Escalate and change root password* `sudo -i`, `echo "root:<NEWPASS>" | chpasswd`.
1–4	*// Run the hardening script (admin user, lock root, SSH hardening, iptables OUTPUT DROP, fail2ban)* `sudo bash harden_vm2_fedora_webmail.sh <NEWPASS> changeme`.
4–6	*// Audit users — mail accounts are likely scored and must NOT be deleted* `sudo bash audit_vm2_fedora_webmail.sh`. **Do NOT delete mail user accounts.**
6–8	*// Verify both mail services are running* `systemctl status postfix` (SMTP) and `systemctl status dovecot` (POP3/IMAP). *// Test SMTP by connecting to port 25* `telnet localhost 25` — should get a 220 banner. *// Test POP3 by connecting to port 110* `telnet localhost 110` — should get +OK.
8–9	*// Verify the webmail UI is accessible* `curl -I http://localhost`. If Roundcube or similar, check it loads.
9–10	*// Check if any mail is stuck in the queue* `mailq`. If stuck: `postfix flush`. *// Check mail logs for errors* `/var/log/maillog` or `/var/log/mail.log`.

Mail Service Troubleshooting

SMTP — Postfix (Port 25 — Scored)

- *// Check/restart Postfix* `systemctl status postfix` / `systemctl restart postfix`
- *// Main Postfix config file* Config: `/etc/postfix/main.cf`
- Key settings: `myhostname`, `mydomain`, `mydestination`, `inet_interfaces = all`
- *// Show all non-default Postfix settings* `postconf -n`
- *// View the mail queue* `mailq`
- *// Force delivery of all queued mail* `postfix flush`
- *// Delete all queued mail (use if the queue is flooded with spam)* `postsuper -d ALL`
- *// Test SMTP manually* `telnet localhost 25` then `EHLO test` — should respond with capabilities
- *// Check mail logs for errors* `tail -f /var/log/maillog` (Fedora) or `/var/log/mail.log`
- *// Verify port 25 is listening* `ss -tlnp | grep :25`
- *// If Postfix won't start, find the config error* **If Postfix won't start:** `postfix check`

POP3 — Dovecot (Port 110 — Scored)

- *// Check/restart Dovecot* `systemctl status dovecot` / `systemctl restart dovecot`
- *// Dovecot config files* Config: `/etc/dovecot/dovecot.conf` and `/etc/dovecot/conf.d/`
- *// Authentication settings* Key file: `/etc/dovecot/conf.d/10-auth.conf`
- *// Mailbox location settings* Key file: `/etc/dovecot/conf.d/10-mail.conf`
- *// Show all non-default Dovecot settings* `doveconf -n`
- *// Test POP3 manually* `telnet localhost 110` then `USER testuser` / `PASS password`
- POP3 scoring uses AD usernames — Dovecot may authenticate against LDAP/AD
- *// Check which auth mechanisms are configured* `grep auth_mechanisms /etc/dovecot/conf.d/10-auth.conf`
- *// Verify POP3 is listening on port 110* `ss -tlnp | grep :110`
- *// If POP3 auth fails, check logs and LDAP/PAM config* **If POP3 auth fails:** check `/var/log/maillog`

Webmail UI (HTTP/HTTPS — Likely Scored)

- Probably Roundcube running on Apache/Nginx with PHP
- *// Check/restart the web server (Fedora uses httpd, not apache2)* `systemctl status httpd` / `systemctl restart httpd`
- *// Web server config location* Config: `/etc/httpd/conf/httpd.conf` , `/etc/httpd/conf.d/`
- *// Roundcube application config* Typically `/var/www/roundcubemail/config/config.inc.php` or similar
- *// Quick test to see if the web server responds* `curl -I http://localhost`
- *// Web server log locations* Logs: `/var/log/httpd/error_log` , `/var/log/httpd/access_log`

Linux Cheat Sheet (Same as Wesley's)

Refer to Wesley's section for the full Linux command reference. Key additions for Fedora:

Fedora-Specific

- *// Fedora uses dnf instead of apt* Package manager: `dnf` (not apt)
- *// Update all packages* `dnf update -y`
- *// Install a package* `dnf install PACKAGE`
- *// Services use systemctl, same as Ubuntu* Services: `systemctl` (same as Ubuntu)
- *// Check if firewalld is running (Fedora default)* Fedora may have `firewalld` active. Check: `systemctl status firewalld`
- *// If firewalld conflicts with iptables rules, disable it* `systemctl stop firewalld && systemctl disable firewalld` then use iptables
- *// Check SELinux enforcement mode* SELinux may be enforcing: `getenforce` . If mail services fail, check: `ausearch -m avc -ts rece`
- *// Fix SELinux denial for webmail connecting to LDAP/DB* **SELinux fix:** `setsebool -P httpd_can_network_connect 1`

Keeping Webmail Running — Priority Checklist

1. Is Postfix running? `systemctl status postfix` → restart if down
2. Is Dovecot running? `systemctl status dovecot` → restart if down
3. Is the web server running? `systemctl status httpd` → restart if down
4. Are ports listening? `ss -tlnp | grep -E ":(25|110|80|443)"`
5. Is the firewall allowing traffic? `iptables -L -n` — verify ports 25, 110, 80, 443 are open for INPUT
6. Is mail being delivered? `mailq` — check for stuck messages
7. Can you authenticate? `telnet localhost 110` → `USER xxx` / `PASS yyy`
8. Check logs: `tail -50 /var/log/maillog`

Red Team Activity Checklist (Every 30–60 Minutes)

1. *// Verify all mail services are running* `systemctl status postfix dovecot httpd`
2. *// Check all scored ports are listening* `ss -tlnp | grep -E ":(25|110|80|443)"`
3. *// Check the mail queue for anomalies (spam flood = red team)* `mailq` — if flooded, `postsuper -d ALL`
4. *// Check for new or unauthorized users* `cat /etc/passwd | awk -F: '$3 >= 1000'`
5. *// Check for suspicious processes*
5. *// Check for new cron jobs* `crontab -l` and `ls -la /etc/cron*`
6. *// Check iptables rules are intact* `iptables -L -n -v` — is OUTPUT still DROP?
7. *// Check mail logs for auth failures or abuse* `tail -50 /var/log/maillog`
8. *// Verify webmail UI is still loading* `curl -I http://localhost`
9. *// Check for recently modified config files* `find /etc -type f -mmin -60`

Quick Reference — All Team

Network Map

```
                    [Internet/Scoring]
                          |
                 [VyOS Router VM11]
                  172.16.101.1  172.16.102.1
                 /                          \
    [PA FW1 VM9]                     [PA FW2 VM10]
Outside: 172.16.101.254       Outside: 172.16.102.254
Inside:  172.20.242.254       Inside:  172.20.240.254
Mgmt:    172.20.242.150       Mgmt:    172.20.240.200
         |                              |
--- Linux LAN ---             --- Windows LAN ---
172.20.242.0/24               172.20.240.0/24
.30  Ubuntu Ecom (VM1)       .102  AD/DNS (VM5)
.40  Fedora Mail (VM2)       .101  Web Server (VM6)
.20  Splunk (VM3)            .104  FTP Server (VM7)
DHCP Ubuntu Wks (VM4)       .100  Win11 Wks (VM8)
```

Scored Services & Responsible Team Member

Service	Port	VM	Owner
HTTP	80	VM1 (Ecom)	Wesley
HTTPS	443	VM1 (Ecom)	Wesley
HTTP	80	VM6 (Win Web)	Gavin
HTTPS	443	VM6 (Win Web)	Gavin
SMTP	25	VM2 (Webmail)	Nathaniel
POP3	110	VM2 (Webmail)	Nathaniel
DNS	53	VM5 (AD/DNS)	Imran
FTP	21	VM7 (FTP)	Imran
TFTP	69	VM7 (FTP)	Imran
NTP	123	VM7 (FTP)	Imran

Critical Rules

- **Password changes on scored accounts → file ticket at auth.ccdc.events FIRST**
- Admin/root password changes do NOT need a ticket
- **Never block entire subnets** on firewalls — /32 only (competition violation)
- Cannot re-task workstations (VM 4, VM 8)
- Cannot change internal IPs or hostnames
- NISE inject portal times are **CST** — we are **EST** (1 hour ahead)
- Inject responses submitted as **PDF** via NISE
- Red team is active the **entire** competition
- SLA penalties for extended service downtime — keep services up!

Default Credentials (Change Immediately!)

VM	Username	Default Password
VM1, VM2, VM4	sysadmin	changeme
VM3 (Splunk)	root / sysadmin	changemenow
VM3 (Splunk Web)	admin	changeme
VM5, VM6, VM7	administrator	!Password123
VM8	administrator / UserOne	!Password123 / ChangeMe123
VM9, VM10 (PA FW)	admin	Changeme123
VM11 (VyOS)	vyos	changeme

Emergency Contacts & Actions

- **Service down?** Check: (1) Is the service process running? (2) Is the port listening? (3) Is the firewall allowing it? (4) Is the Palo Alto rule correct?
- **Red team IP found?** Tell Gray AND Gavin — block /32 on BOTH firewalls immediately.
- **Scored account locked out?** Unlock it. *// Unlock on Windows* `Unlock-ADAccount` or *// Unlock on Linux* `passwd -u USER`.
- **Web content defaced?** Restore from backup. *// Find recently modified web files* `find /var/www -mtime -1` or check IIS logs.